Transistor-Level LDO and PDN-Aware Two-Phase Buck in 130 nm CMOS

Circuit design, transient-assist tradeoffs, and reproducible numerical verification

Abstract - Two complementary power-delivery studies are presented. A nine-MOS analog LDO regulates 0.9 V from 1.2 V and is evaluated over 164 static/DC conditions and ten transient runs. A mixed-level two-phase Buck combines transistor power bridges, drivers and window detectors with behavioral PI/PWM control. Forty-seven main cases and eight gate-resistance cases characterize remote-load droop, unload overshoot and solver sensitivity. Complete device sizing and block functions connect circuit decisions to measured simulation outcomes. All results are pre-layout; the reference, passive-network assumptions and controller-power boundary are stated with the metrics.

I. DESIGN OBJECTIVE AND VERIFICATION

Power delivery is assessed at the point where the load receives energy. Regulation accuracy, response latency, capacitance, parasitic impedance and control effort are evaluated together. The two circuits are separate studies; this report does not imply that they have already been integrated into a hierarchical supply system.

The analysis distinguishes transistor simulation, saved numerical samples, derived measurements and normalized metrics. A case finishing successfully establishes data availability, while stability margin, waveform recovery and device headroom require their own evidence.

A. LDO regulation path

M2-M3 sense VREF and the divided output. M4 supplies the input-stage tail current; M5 and the 100-kohm resistor establish bias. M6-M7 form the mirror load. The M8-M9 common-source driver controls the gate of PMOS pass device M1. The feedback divider is 200 kohm / 200 kohm. The dominant series compensation branch uses 36 kohm and 18 pF.

Negative feedback was checked from saved connectivity and DC operating points. Increased feedback voltage reduces EAOUT, raises the pass gate through the driver and reduces pass conduction. The STB probe between FB and FBIN is included in the detailed circuit description.

B. Buck energy and control paths

The two power bridges are driven 180 degrees apart. Each phase supplies a 2.2-uH external inductor. VCONV is connected to the remote VLOAD through declared resistance and inductance. Output and load capacitors include the stated ESR/ESL test parasitics.

The transistor implementation contains 12 power-bridge MOS, 24 gate-driver MOS, 20 voltage-window-comparator MOS and three source/sink-assist MOS instances. The controller samples voltage and ideal current-sense signals, generates the PI/PWM commands and supervises the finite assist sequence. Its internal power is not modeled.

The selected assist strategy changes the PI current state by 40 mA while the analog helper sources or sinks current at VLOAD. The arbitration interval is 2 us; its end does not erase the resulting integrator state. Gate resistance in revision r9 modifies assist edge rates while preserving device geometry.

TABLE I. PRINCIPAL TEST CONDITIONS

Quantity	LDO	Two-phase Buck
Process / MOS instances	130 nm / 9	130 nm / 59
VIN / target VOUT	1.2 / 0.9 V	1.8 / 0.9 V
Nominal load step	0.1 to 5 mA	10 to 100 mA
Test capacitance	External 1 nF	1 uF + remote 10 nF
Switching frequency	Continuous-time	5 MHz nominal
Reference / control	Ideal / transistor	Behavioral PI/PWM

Figures 1-2 identify each functional path. Section III defines the normalized metrics and their denominators. Appendix A discloses unit width, length, parallel multiplicity, finger count and total drawn width.

II. CIRCUIT ARCHITECTURES

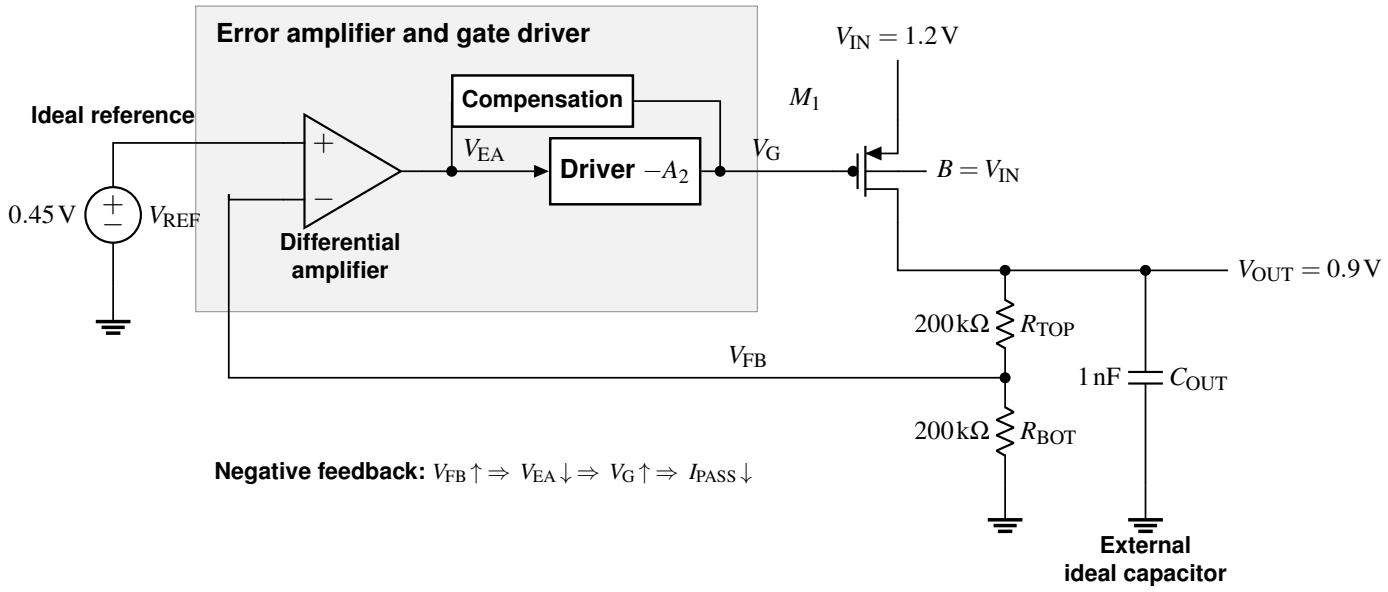


Fig. 1. LDO regulation path, bias, feedback and compensation. Solid power connections and explicit feedback polarity identify the implemented loop. The reference and external output capacitor are testbench elements.

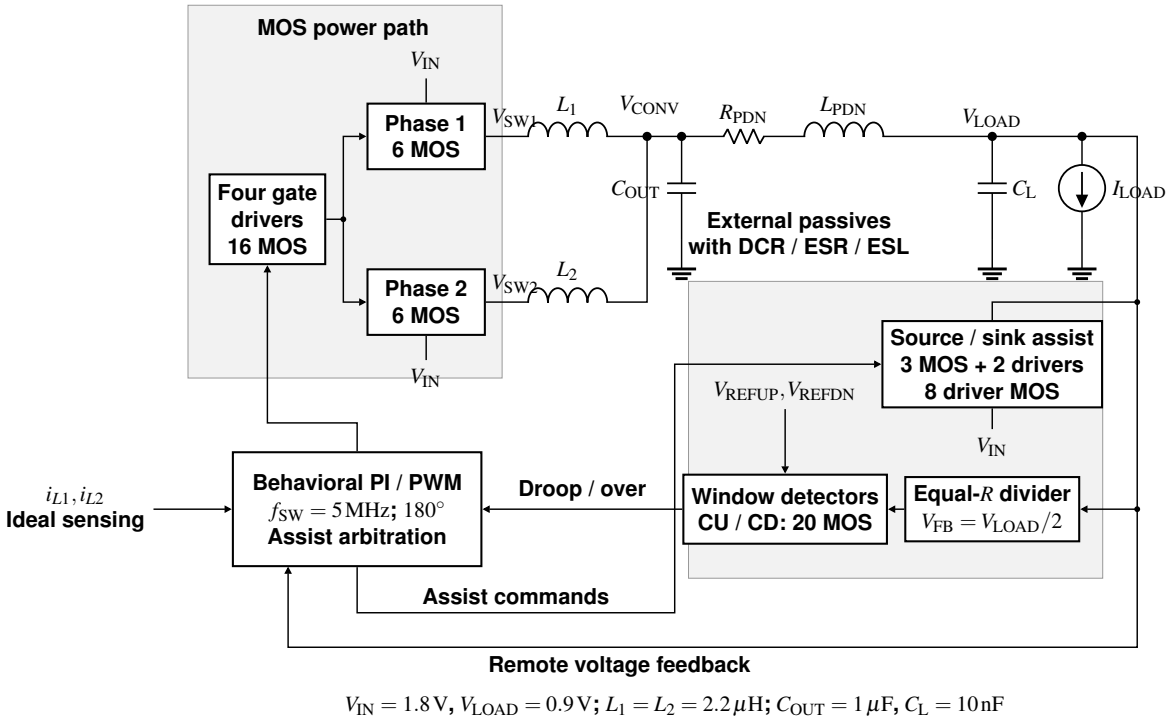


Fig. 2. Two-phase Buck, remote load and transient-assist architecture. The energy path, analog detector path and behavioral control boundary are distinguished. Source and sink assist act at VLOAD.

III. FIGURES OF MERIT AND MEASUREMENT DEFINITIONS

A. Capacitance-weighted LDO FoM

The primary scalar follows the sum-excursion convention in [1, Eq. 14]. Lower is better only when the operating conditions and metric boundaries are comparable.

$$F_{C,sum,max} = C_{ext} (dV_{under} + dV_{over}) I_Q / I_{max}^2 \text{ [s]}$$

Here Cext = 1 nF is an external ideal zero-ESR test capacitor; Imax = 5 mA; the load step is 0.1 to 5 mA. Voltage excursions are measured against the fixed 0.9-V target. IQ is the larger of the matched light/heavy DC endpoints at the same process, temperature, input voltage and revision. It includes the modeled amplifier, bias, driver and divider; implemented-reference power is absent.

TABLE II. PRIMARY LDO FoM

Condition	IQ (uA)	Sum (mV)	FoM (ps)
TT, 27 C, 100 ns	70.162	244.543	686.31
TT, 27 C, 10 ns	70.162	253.328	710.96
SS, 125 C, 100 ns	71.569	322.796	924.08

B. Separate per-edge convention

The step-based single-spike definition [2, Table I] is reported separately:

$$F_{C,edge,step} = C_{ext} dV_{edge} I_Q / (dI_{load})^2 \text{ [s]}$$

At TT, 27 C and 100 ns, dI = 4.9 mA gives 370.60 ps for load-up and 344.00 ps for unload. These numbers use one directional peak and a different denominator from Table II.

Cext dVunder / dI = 25.882 ns is a charge-deficit time proxy. Actual load-up recovery is 1.690 us. The proxy is not a settling-time or bandwidth measurement.

C. Buck normalized transient diagnostics

Two transparent descriptors are used because physical controller power and active layout area are unavailable:

Normalized excursion = 100 dVpeak / Vtarget [%]
Recovery cycles = fsw $t_{recovery}$ [cycles]

fsw is the per-phase switching frequency, 5 MHz for the selected cases. OFF/ON comparisons require the same PDN, passives, load edge, phase offset and solver settings, except for the explicitly changed assist flags. They describe this design and do not establish a universal Buck ranking.

Condition	OFF peak (%)	ON peak (%)
Nominal up	15.930	11.932
Hard-PDN down	9.654	12.255

Nominal load-up recovery changes from 37.736 to 68.703 cycles. A smaller peak therefore coexists with slower recovery. The hard-PDN unload case also exposes an increased overshoot.

D. Recovery and energy boundaries

Recovery uses the 0.891-0.909 V band and is qualified only through the finite observation window. LDO recovery is timed from the ramp end. Buck raw-grid and cycle-mean recovery are timed from the event start and remain separate measurements.

Helper work is the signed integral of VLOAD times helper current over the declared event window. Source and sink directions are retained separately. Modeled supply-port event energy includes converter, driver and helper interactions; its OFF/ON difference is not helper-only energy or full physical efficiency.

REFERENCES FOR METRIC CONVENTIONS

[1] Y. Zhang et al., output-capacitor-less LDO with sub-threshold slew-rate enhancement, *Micromachines*, vol. 15, art. 1019, 2024, Eq. 14. doi:10.3390/mi15081019.
[2] Y. Yao et al., *IEICE Electronics Express*, vol. 19, art. 20220051, 2022, Table I. doi:10.1587/elex.19.20220051.

IV. LDO PERFORMANCE AND STABILITY BOUNDARY

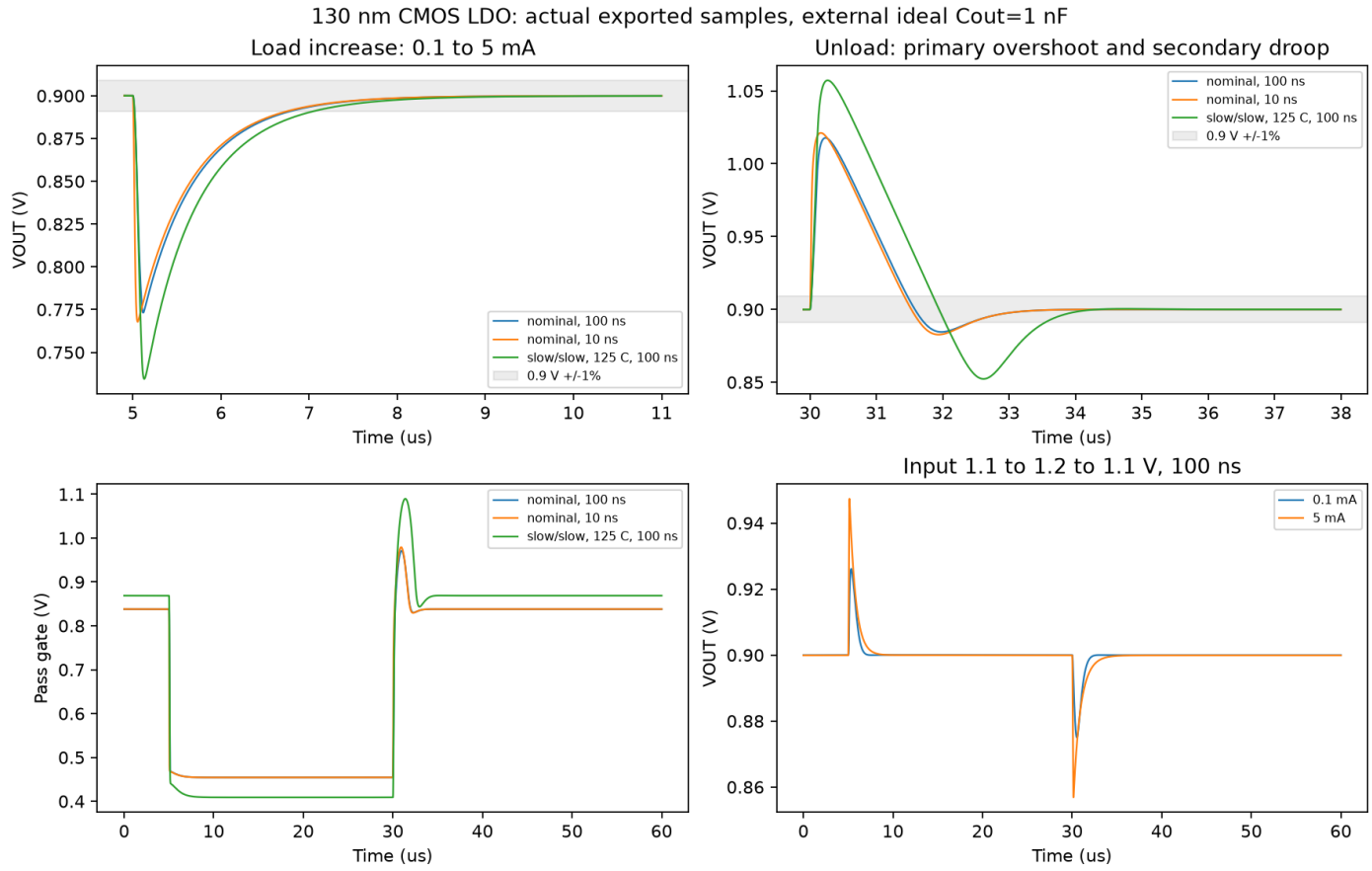


Fig. 3. Load-step response, pass-gate response and input-step response from exported simulation samples. The ideal external output capacitance is 1 nF. Curves include the nominal and slow/hot conditions.

Condition	Up under (mV)	Down over (mV)	Up recovery (us)	Down recovery (us)	Secondary under (mV)
TT, 27 C, 100 ns	126.823	117.720	1.690	2.318	15.472
TT, 27 C, 10 ns	132.183	121.145	1.742	2.419	17.295
SS, 125 C, 100 ns	165.554	157.242	1.954	3.439	47.707

A. Stability versus test capacitance

At TT, 27 C and 1 nF, native loop results at 0.1 / 1 / 5 mA give phase margins of 72.44 / 76.17 / 74.76 degrees and gain margins of 38.46 / 23.19 / 18.47 dB. At 100 pF the 5-mA margin falls to 6.62 degrees / 1.27 dB. All three 10-pF cases are identified as unstable by the native analysis.

The 1-nF compensation result does not establish capacitor-less operation. Vin = 1.1-V corners also reveal limited tail/driver headroom; saturation is assessed per operating point.

B. DC accuracy and numerical checks

The TT load-regulation slope is -0.01185 mV/mA over 0.1-5 mA. Line-regulation slopes are 0.3753 / 0.4766 mV/V at light/heavy load over Vin = 1.1-1.2 V. The 5-mA dropout estimate is about 131 mV under the stated 1% definition, bracketed by a 10-mV input sweep.

Independent maxstep comparisons change the selected primary peaks by at most 11.864 uV. PSRR, integrated output noise, PVT, full event definitions and all smaller-capacitance failures remain available in the repository. Noise excludes implemented-reference noise.

V. BUCK TRANSIENT-ASSIST TRADEOFFS

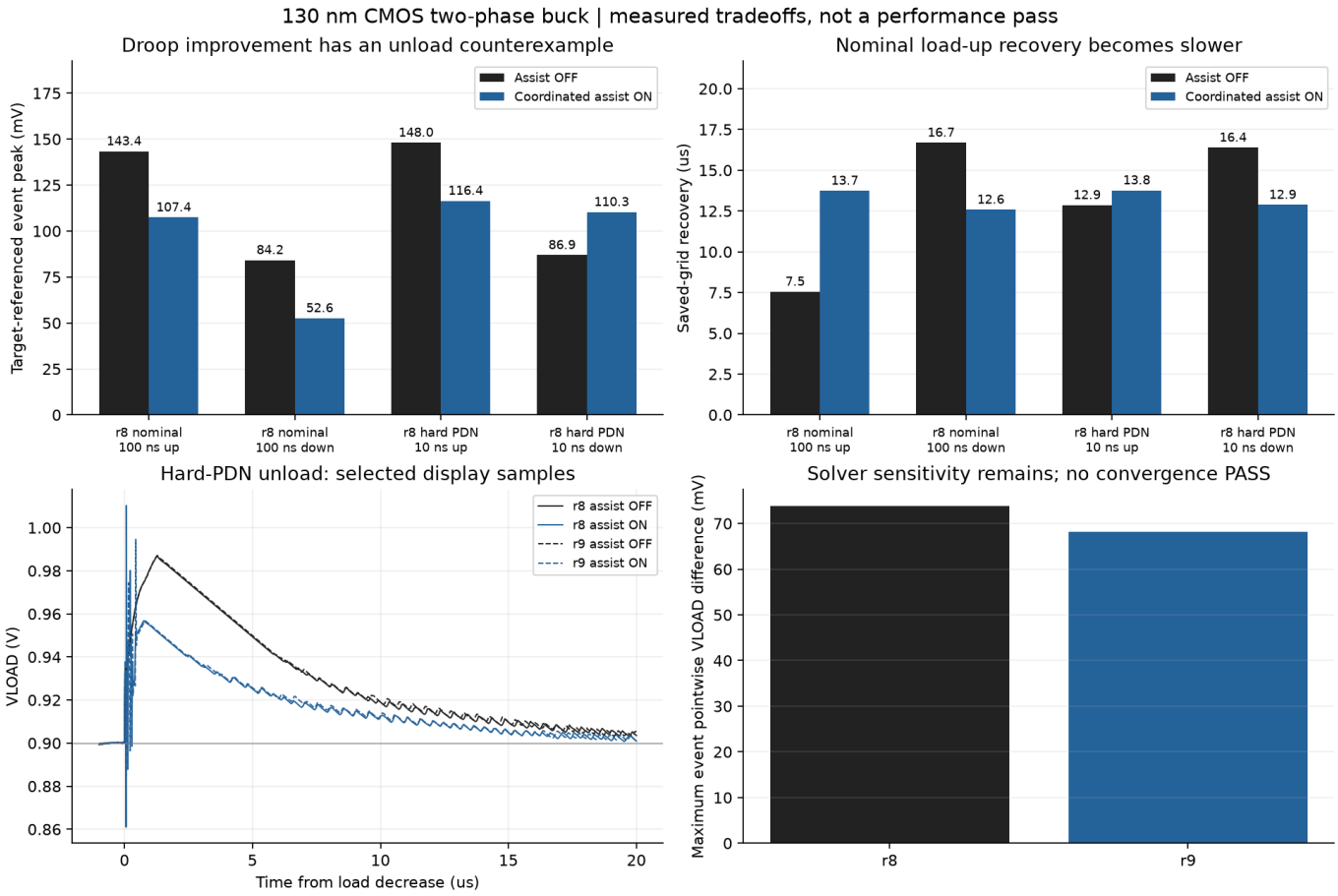


Fig. 4. Matched OFF/ON transient outcomes, hard-PDN unload behavior and solver sensitivity. The waveform panel uses selected display samples; scalar measurements and event work retain their original full-grid measurement provenance.

A. Assist benefit and adverse conditions

At nominal 100-ns load-up, coordinated assist reduces droop from 143.370 to 107.386 mV but increases recovery from 7.547 to 13.741 us. Nominal unload overshoot improves from 84.175 to 52.562 mV. These two directions have different energy and control-state demands.

For RPDN = 0.3 ohm, LPDN = 2 nH and CLOAD = 2 nF with a 10-ns edge, r8 unload overshoot rises from 86.889 to 110.298 mV. Revision r9 gate resistance reduces the ON peak to 94.554 mV; its matched OFF result is 87.129 mV. The adverse condition remains visible.

B. Analog and behavioral partition

The transistor detectors, bridge, gate drivers and helper devices establish circuit-level interactions. PI/PWM, timing, supervision, programmable reference and ideal current sensing are behavioral. Port energy is meaningful within that modeled boundary; full physical efficiency and controller-IQ FoMs are unavailable.

C. Coverage and solver sensitivity

The main set has 47 completed cases, 24 declared comparisons and 94 events. All 141 steady means fall within the target band, but 127 finite-window periodicity checks are not accepted. The r9 set adds eight cases and four matched comparisons. Completion and average regulation do not establish switching-loop signoff.

Main-set sensitivity checks show up to 1.245-mV primary-peak change and 73.770-mV pointwise VLOAD difference. For r9, the hard-PDN ON unload peak changes from 94.554 to 97.637 mV in the finer solve. Recovery and ringing remain solver-sensitive.

D. Reproducibility and scope

The public scripts regenerate numerical comparisons and figures from the published exports. Hashes bind inputs and curated assets. Complete sizing, terminal ordering, assist supervision and passive values are provided in each project directory.

All reported results are schematic-level simulations. Layout extraction, mismatch/yield, absolute-maximum qualification, hardware measurements and a closed switching-loop stability analysis remain future verification work.

APPENDIX A. COMPLETE MOS SIZING

W and L are in micrometers. Wtot = Wunit x m x nf; all reported nf values are 1. These are drawn transistor dimensions, not layout area. Drain/gate/source/bulk connections and block assignments are provided in the repository design tables.

LDO: all nine MOS instances

Device	P/N	Wunit	L	m	nf	Wtot
M1	P	10	0.15	10	1	100
M2	P	2.5	1	4	1	10
M3	P	2.5	1	4	1	10
M4	P	2.5	1	8	1	20
M5	P	2.5	1	2	1	5
M6	N	2	1	2	1	4
M7	N	2	1	2	1	4
M8	P	2.5	1	8	1	20
M9	N	2	1	4	1	8

Buck: instances 1-25 of 59

Device	P/N	Wunit	L	m	nf	Wtot
CD_NI1	N	1	0.15	1	1	1
CD_NI2	N	1	0.15	1	1	1
CD_NL	N	2	1	4	1	8
CD_NR	N	2	1	4	1	8
CD_PB	P	2.5	1	1	1	2.5
CD_PI1	P	2	0.15	1	1	2
CD_PI2	P	2	0.15	1	1	2
CD_PL	P	5	0.5	4	1	20
CD_PR	P	5	0.5	4	1	20
CD_PT	P	2.5	1	16	1	40
CU_NI1	N	1	0.15	1	1	1
CU_NI2	N	1	0.15	1	1	1
CU_NL	N	2	1	4	1	8
CU_NR	N	2	1	4	1	8
CU_PB	P	2.5	1	1	1	2.5
CU_PI1	P	2	0.15	1	1	2
CU_PI2	P	2	0.15	1	1	2
CU_PL	P	5	0.5	4	1	20
CU_PR	P	5	0.5	4	1	20
CU_PT	P	2.5	1	16	1	40
DAD_N1	N	4	0.5	2	1	8
DAD_N2	N	10	0.5	20	1	200
DAD_P1	P	8	0.5	2	1	16
DAD_P2	P	20	0.5	20	1	400
DAU_N1	N	4	0.5	2	1	8

Buck: instances 26-59 of 59

Device	P/N	Wunit	L	m	nf	Wtot
DAU_N2	N	10	0.5	20	1	200
DAU_P1	P	8	0.5	2	1	16
DAU_P2	P	20	0.5	20	1	400
DH1_N1	N	4	0.5	2	1	8
DH1_N2	N	10	0.5	20	1	200
DH1_P1	P	8	0.5	2	1	16
DH1_P2	P	20	0.5	20	1	400
DH2_N1	N	4	0.5	2	1	8
DH2_N2	N	10	0.5	20	1	200
DH2_P1	P	8	0.5	2	1	16
DH2_P2	P	20	0.5	20	1	400
DL1_N1	N	4	0.5	2	1	8
DL1_N2	N	10	0.5	20	1	200
DL1_P1	P	8	0.5	2	1	16
DL1_P2	P	20	0.5	20	1	400
DL2_N1	N	4	0.5	2	1	8
DL2_N2	N	10	0.5	20	1	200
DL2_P1	P	8	0.5	2	1	16
DL2_P2	P	20	0.5	20	1	400
NLS1	N	10	0.5	100	1	1000
NLS1_B1	N	10	0.5	100	1	1000
NLS2	N	10	0.5	100	1	1000
NLS2_B1	N	10	0.5	100	1	1000
PASSDN	N	10	0.5	60	1	600
PASSUP	P	10	0.5	60	1	600
PASSUP_B1	P	10	0.5	60	1	600
PHS1	P	10	0.5	100	1	1000
PHS1_B1	P	10	0.5	100	1	1000
PHS1_B2	P	10	0.5	100	1	1000
PHS1_B3	P	10	0.5	100	1	1000
PHS2	P	10	0.5	100	1	1000
PHS2_B1	P	10	0.5	100	1	1000
PHS2_B2	P	10	0.5	100	1	1000
PHS2_B3	P	10	0.5	100	1	1000

Revision r9 retains these 59 MOS geometries and adds 1-kohm / 2-kohm assist gate resistors. Some driver drain node names change in r9; its separate connectivity inventory records those changes.